
```

n = 5;
x_exact = (n:-1:1)';
A_gauss = rand(n, n);
b_gauss = A_gauss * x_exact;
sol_gauss = gausse(A_gauss, b_gauss);
disp(sol_gauss);

function solutie = gausse(MatA, termen_b)
    dim = length(termen_b);
    MatExt = [MatA, termen_b]; %matricea

    for k = 1:dim-1

        [~, max_relativ] = max(abs(MatExt(k:dim, k)));
        linie_pivot = max_relativ + k - 1;

        if abs(MatExt(linie_pivot, k)) < eps
            error('sistemul este singular.');
```

end

```

        if linie_pivot ~= k
            MatExt([k, linie_pivot], :) = MatExt([linie_pivot, k], :);
        end

        for rand_curent = k+1:dim
            multiplicator = MatExt(rand_curent, k) / MatExt(k, k);
            MatExt(rand_curent, k:end) = MatExt(rand_curent, k:end) -
multiplicator * MatExt(k, k:end);
        end
    end

    if abs(MatExt(dim, dim)) < eps
        error('sistemul este singular .');
```

end

```

    solutie = zeros(dim, 1);
    solutie(dim) = MatExt(dim, dim+1) / MatExt(dim, dim);

    for k = dim-1:-1:1
        solutie(k) = (MatExt(k, dim+1) - MatExt(k, k+1:dim) *
solutie(k+1:dim)) / MatExt(k, k);
    end

end

5.0000
4.0000
3.0000
2.0000
```

1.0000

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